Chapter 15. The Derivatives in n-Space

1. Find the **domain** and **range** of a given function $f(x,y)$ and sketch the graph $z = f(x,y)$ (Page 634).

2. Sketch the level curves (contour map) $z = k$ in the plane for a given function $z = f(x,y)$ with indicated values of k (Page 635-638).

3. Given a function $f(x,y)$, find all its **first partial derivatives**

$$f_x(x,y) = \frac{\partial f(x,y)}{\partial x}, \quad f_y(x,y) = \frac{\partial f(x,y)}{\partial y}$$

and four **second partial derivatives** $f_{xx}, f_{xy}, f_{yx}, f_{yy}$. Given a function $f(x,y,z)$, find all its first partial derivatives f_x, f_y, f_z.

4. Find the indicated limit of a given function or the state if it doesn't exit.

5. Difinitions for **local linearity** and **differentibility** (Page 651-652). For a function $f(x,y)$, the vector

$$\nabla f(\mathbf{p}) =< f_x(\mathbf{p}), f_y(\mathbf{p}) >= f_x(\mathbf{p})\mathbf{i} + f_y(\mathbf{p})\mathbf{j}$$

is called **gradient** of f. For a function $f(x,y,z)$,

$$\nabla f(\mathbf{p}) =< f_x(\mathbf{p}), f_y(\mathbf{p}), f_z(\mathbf{p}) >= f_x(\mathbf{p})\mathbf{i} + f_y(\mathbf{p})\mathbf{j} + f_z(\mathbf{p})\mathbf{k}$$

For properties of ∇, see Theorem B in Page 654.

6. Let $\mathbf{p} = (x,y)$ and $\mathbf{p}_0 = (x_0, y_0)$, the equation

$$z = T(\mathbf{p}) = f(\mathbf{p}_0) + \nabla f(\mathbf{p}_0) \cdot (\mathbf{p} - \mathbf{p}_0)$$

defines a plane that approximates f near $\mathbf{p}_0$ and it is called the **tangent plane** of f at $\mathbf{p}$ (Page 653).

7. The directional derivative of $f(x,y)$ at $\mathbf{p} = (x,y)$ in the direction of unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is given by:

$$D_{\mathbf{u}} f(\mathbf{p}) = \mathbf{u} \cdot \nabla f(\mathbf{p}) = u_1 f_x(x,y) + u_2 f_y(x,y)$$

8. Maximum rate of change, see Theorem B in page 657.

9. **Chain Rules** (Page 661-663):

- If $x = x(t)$, $y = y(t)$, $z = f(x,y)$, then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

- If $x = x(s,t)$, $y = y(s,t)$, $z = f(x,y)$, then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

10. Implicit differentiation (Page 664):

- Suppose that $F(x, y) = 0$ define y implicitly as a function of x, then

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}$$

- Suppose that $F(x, y, z) = 0$ define z implicitly as a function of x and y, then

$$\frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z}, \quad \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}$$

11. For the surface $F(x, y, z) = k$, the euqation of tangent plane at (x_0, y_0, z_0) is given by

$$\nabla F(x_0, y_0, z_0) \cdot < x - x_0, y - y_0, z - z_0 > = 0,$$

i.e.,

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

See Theorem A in Page 667.

12. The total differential of a function $f(x, y)$ is defined by

$$dz = df(x, y) = f_x(x, y)dx + f_y(x, y)dy.$$

13. Definitions of global (local) maximum and minimum values, critical points (Page 671-672) and saddle points for a function f.

14. Find the extreme values for a given function f using the **second partial test** (Theorem C in Page 672).

15. Maximize or minimize a function $f(\mathbf{p})$ with constraint $g(\mathbf{p}) = 0$ using **Lagrange's method** (Theorem A in Page 678). Lagrange's method is also called *method of Lagrange multipliers.*

16. Maximize or minimize a function $f(x, y, z)$ with constraints

$$g(x, y, z) = 0, \quad h(x, y, z) = 0$$

using Lagrange's method (Page 680).

Solution process for extreme value problems

Problem 1-a. Find critical points of a function $f(x,y)$.

SOLUTION. To find the critical points, we need to solve the equations

$$f_x(x,y)=0 \quad f_y(x,y)=0 \quad \square$$

Problem 1-b. Find critical points of a function $f(x,y,z)$.

SOLUTION: To find the critical points, we need to solve the equations

$$f_x(x,y,z)=0 \quad f_y(x,y,z)=0 \quad f_z(x,y,z)=0 \quad \square$$

Problem 2. Find the local extreme values and saddle point(s) of a function $f(x,y)$.

SOLUTION:
STEP 1. Find the critical points (see Problem 1).
STEP 2. Apply the second derivatives test (Theorem C in Page 672). $\square$

Problem 3. Find the maximum and minimum values of $f(x,y,z)$ subject to the constraint $g(x,y,z)=0$ without using Lagrange multipliers.

SOLUTION:
STEP 1. Solve the equation $g(x,y,z)=0$ for x, y, or for z, and substitute the obtained expression into $f(x,y,z)$. Assume that we solve $g(x,y,z)=0$ for z, then $z=z(x,y)$ and

$$f(x,y,z(x,y))=h(x,y)$$

STEP 2. Apply Problem 2 to the function $h(x,y)$. $\square$

For example, if $f=xyz$ and $g(x,y,z)=x+2y+z-4=$, we can substitute $z=4-x-2y$ into f, and so we have

$$f(x,y,z)=xy(4-x-2y)=h(x,y)$$

Problem 4. Use *Lagrange's method* to find the maximum and minimum values of $f(x,y,z)$ subject to the constraint $g(x,y,z)=0$.

SOLUTION:
STEP 1. Solve the equations

$$\begin{aligned}
&f_x(x,y,z)=\lambda g_x(x,y,z)\\
&f_y(x,y,z)=\lambda g_y(x,y,z)\\
&f_z(x,y,z)=\lambda g_z(x,y,z)\\
&g(x,y,z)=0
\end{aligned}$$

STEP 2. Evaluate f at all points (x,y,z) obtained in the first step. The largest of these values is the maximum value of f; the smallest is the minimum value of f. $\square$

Problem 5. Use *Lagrange's method* to find the maximum and minimum values of $f(x, y, z)$ subject to the given constraints $g(x, y, z) = 0$ and $h(x, y, z) = 0$

SOLUTION:

STEP 1. Solve the equations

$$\begin{aligned}
&f_x(x, y, z) = \lambda g_x(x, y, z) + \mu h_x(x, y, z) \\
&f_y(x, y, z) = \lambda g_y(x, y, z) + \mu h_y(x, y, z) \\
&f_z(x, y, z) = \lambda g_z(x, y, z) + \mu h_z(x, y, z) \\
&g(x, y, z) = 0 \\
&h(x, y, z) = 0
\end{aligned}$$

STEP 2. Evaluate f at all points (x, y, z) obtained in the first step. The largest of these values is the maximum value of f; the smallest is the minimum value of f. $\square$